

Brian Kyanjo

Computational Scientist | Scientific Software Developer

HPC & Scientific Computing | Numerical Modeling | Open-Source Research Software

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Professional Summary

Computational scientist and scientific software developer specializing in high-performance scientific computing, numerical modeling, data assimilation, and scalable research software. My work focuses on developing reliable, reproducible, and interoperable computational systems for large-scale scientific applications. I have experience developing open-source scientific software, MPI- and GPU-based workflows, HPC and cloud computing platforms, containerized environments, and coupled modeling systems across Python, MATLAB, and compiled scientific codes.

Education

Boise State University (BSU)

2024

Computing Ph.D. in Computational Math Sciences and Engineering

Dissertation: *GeoFlood: A Computational Model for Overland Flooding*

Advisor: Prof. Donna Calhoun

African Institute for Mathematical Sciences (AIMS), Rwanda

2020

Master of Science in Mathematical Sciences (Climate Resilience)

Thesis: *Simulating Tropical Cyclone over the Southwest Indian Ocean using Model for Prediction Across Scales (MPAS): Sensitivity to Cloud Microphysics Schemes*

Advisor: Prof. Babatunde J. Aboidun

Makerere University (MAK)

2018

Bachelor of Science in Physical Sciences (Physics and Mathematics)

Final Project: *Water Level Sensor System (Design and Construction)*

Advisor: Prof. Ireeta Tumps Winston

Technical Skills

Scientific Computing: Numerical PDEs, finite volume methods, adaptive mesh refinement (AMR), data assimilation, ensemble methods, uncertainty quantification, numerical modeling

Programming: Python, C++, C, FORTRAN, MATLAB, CUDA, R, Java, Bash

HPC and Parallel Computing: MPI, OpenMP, CUDA, Slurm, distributed-memory computing, GPU acceleration, parallel I/O, scalable scientific workflows

Scientific Software Engineering: Git, Linux, APIs, model coupling, Python–MATLAB interoperability, testing, workflow automation, reproducible computational workflows

Research Infrastructure: Docker, Apptainer, Spack, Jupyter, AWS/cloud computing, HPC deployment, containerized scientific environments

Additional: L^AT_EX, SQL, HTML, electronics/auto technician experience

Professional Experience

Georgia Institute of Technology

Sept. 2026 – Present

Research Scientist II

Previous appointment: Postdoctoral Research Fellow – Computational Science and Scientific Software, Aug. 2024 – Aug. 2026.

- Design and develop ICESEE, an open-source, MPI-parallel ensemble data assimilation framework for state and parameter estimation in ice-sheet and geophysical models.
- Develop scalable model-coupling interfaces and workflows for integrating ICESEE with independent scientific models, including ISSM and Icepack.
- Parallelize and optimize ensemble simulation and data assimilation workflows for distributed-memory HPC systems using MPI and scalable scientific computing methods.
- Design and develop CryoStack, a scientific computing platform for configuring, executing, monitoring, and managing reproducible scientific workflows across institutional HPC and cloud computing environments.
- Develop Remote and AWS cloud execution capabilities for scientific workflows, including containerized execution on AWS Fargate and EC2 and integration with institutional computing resources.
- Develop reproducible scientific software environments using Docker, Apptainer, Spack, and HPC deployment workflows.
- Develop Python–MATLAB interoperability and ISSM–ICESEE coupling workflows for ensemble forecasting, parameter estimation, and data assimilation.
- Develop workflow automation for experiment configuration, provenance, simulation diagnostics, HPC job orchestration, and scientific result management.
- Contribute to research on scalable data assimilation and computational methods for understanding ice-sheet dynamics and their contribution to future sea-level change.

Georgia Institute of Technology

Apr. 2025 – Present

Research Faculty Advisory Council (RFAC), EAS

- Represent research faculty interests within the College of Sciences.
- Serve as a liaison between research faculty and college leadership.
- Advance initiatives supporting faculty engagement, inclusion, and professional growth.
- Evaluate internal grant proposals and contribute to funding decisions.

Rocky Mountain Advanced Computing Consortium (RMACC)

Oct. 2023 – Sept. 2024

RMACC Apprenticeship

- Designed, quoted, and deployed a major expansion of the regional high-performance computing cluster at the University of Colorado Boulder (Alpine supercomputer).
- Designed a research computing cluster in a Unix environment.

- Supported HPC cluster provisioning, deployment automation, and systems administration workflows.
- Created playbooks and container images for MPI-based applications.

Boise State University, School of Computing
Graduate Teaching Assistant

June 2023 – Dec. 2023

- Developed a computational model for overland flooding: github.com/KYANJO/GeoFlood.
- Developed a CUDA-based version of GeoFlood.
- Served as teaching assistant for Introduction to Programming in Mathematics.

Boise State University, School of Computing
Graduate Research Assistant

Aug. 2020 – July 2024

- Developed GeoFlood, a computational model for overland flooding: github.com/KYANJO/GeoFlood.
- Benchmarked GeoFlood across CPU and GPU computing platforms.
- Developed CUDA-based Riemann solvers for GeoFlood.
- Developed ForestClaw extensions for atmospheric applications.
- Explored data-driven and machine learning approaches to enhance predictive modeling and simulation accuracy.
- Implemented a discontinuous Galerkin solver within ForestClaw as a summer research project.
- Completed a comprehensive examination on Riemann solvers for wet/dry interfaces in finite-volume schemes: github.com/KYANJO/Comp_exam/tree/main/code/comp_exam_artifact.
- Implemented approximate and exact Riemann solvers for handling wet/dry interfaces using finite-volume schemes.
- Investigated accelerated wave-propagation algorithms using GPU and CPU comparisons on HPC platforms.

Idaho National Laboratory (INL)
Intern

May 2022 – Sept. 2022

- Supported post-irradiation examination capabilities for nuclear fuel using gamma tomography.
- Simulated gamma-emission tomography of nuclear fuel and generated tomographic reconstructions to identify isotopic distributions.
- Performed thermal-fluid analyses of geothermal closed-loop configurations and discussed results with the INL research team.
- Participated in team meetings and contributed to maintenance and improvement of the software suite.

AIMS Rwanda
Tutor

Sept. 2020 – Dec. 2020

- Tutored master's-level students in mathematics.
- Assisted lecturers with academic quality assurance.
- Supported administrative, library, IT, logistics, and technical activities as needed.

ESMT Berlin
Certificate in Business Analytics

2020

Marga Business Simulations, Germany
Certificate in Business Simulations Management

2020

- Developed and marketed products within a competitive business simulation.
- Collaborated with six students from different universities and disciplines.
- Served as team lead.

God Cares High School
Physics and Mathematics Teacher

Jan. 2020 – July 2020

- Taught physics and mathematics.
- Supervised laboratory activities.

Ubuntu Hill School
Physics and Mathematics Teacher

Aug. 2019 – July 2020

- Taught physics and mathematics.

Kampala Capital City Authority (K.C.C.A.)
Intern

May 2017 – Aug. 2017

- Fabricated, repaired, and replaced damaged street lights and electronic control boxes.
- Troubleshoot, maintained, and installed electric and solar street lights across the city.
- Supported maintenance of traffic lights and investigated approaches for reducing peak-hour traffic.

Selected Funded Projects

- **Arête Glacier Initiative Sliding Law RFP**, *A Framework for Detection of Changing Subglacial Properties through Transient Data Assimilation*, 2026–2028.
Award: \$186,965. PI: Alexander Robel; Co-PI: Talea Mayo; Co-Investigator: Brian Kyanjo. The project focuses on transient data assimilation for detecting changes in subglacial properties and improving constraints on future sea-level projections.

Selected Scientific Software

- **ICESEE (Ice Sheet State and Parameter Estimator)** – Open-source, MPI-parallel ensemble data assimilation framework for state and parameter estimation in ice-sheet and geophysical models.
github.com/ICESEE-project/ICESEE
- **CryoStack** – Scientific computing platform for configuring, executing, monitoring, and managing reproducible scientific workflows across institutional HPC and AWS cloud environments.
github.com/ICESEE-project/CryoStack
- **ICESEE-Containers** – Reproducible container infrastructure for deploying MPI-based scientific software across HPC systems using Docker and Apptainer.
github.com/ICESEE-project/ICESEE-Containers

- **ICESEE-Spack** – Spack-based software environment for reproducible builds and deployment of ICESEE and coupled scientific modeling stacks.
github.com/ICESEE-project/ICESEE-Spack
- **GeoFlood** – Hybrid CPU/GPU computational model for overland flooding using finite volume methods and adaptive mesh refinement.
github.com/KYANJO/GeoFlood
- **ForestClaw Extensions** – Numerical solver and adaptive mesh refinement development for PDE-based geophysical simulations.
github.com/KYANJO/forestclaw

Journal Publications and Refereed Conference Proceedings

- Brian Kyanjo and Alexander Robel. *The Ice Sheet State and Parameter Estimator (ICESEE) Library: Ensemble Kalman Filtering for Ice Sheet Models*. Submitted. 10.48550/arXiv.2603.26947.
- Brian Kyanjo, Donna Calhoun, and David L. George. *A Hybrid CPU/GPU GeoFlood: High-Performance Computing in Overland Flooding*. In preparation.
- Brian Kyanjo, Donna Calhoun, and David L. George. *GeoFlood: Computational Model for Overland Flooding*. Submitted. 10.48550/arXiv.2403.15435.
- B. Kyanjo, D. Calhoun, C. Burstedde, S. Aiton, J. Snively, and M. Shih. *GPU Accelerated Adaptive Wave Propagation Algorithm*. 2023. 10.13140/RG.2.2.11304.34569.
- Carlo Parisi, Paolo Balestra, Brian Kyanjo, Theron D. Marshall, Travis L. McLing, and Mark D. White. *Closed Loop Geothermal Analysis Modeling and Simulation Using Idaho National Laboratory's RELAP5-3D-FALCON Coupled Codes*. Proceedings of the 48th Workshop on Geothermal Reservoir Engineering, Stanford University, Stanford, California, February 6–8, 2023, SGP-TR-224.

Selected Conferences and Short Courses

- American Geophysical Union (AGU) Annual Meeting, 2025
- Society for Industrial and Applied Mathematics (SIAM), 5th Biennial Meeting of the SIAM Pacific Northwest Section, 2025
- Scalable and Computationally Reproducible Approaches to Arctic Research, 2025
- Adviser and Icepack Workshop and Hackathon, 2025
- The International Conference for High Performance Computing, Networking, Storage, and Analysis (SC), 2024
- United States Research Software Engineers (US-RSE) Conference, *Software-Enabled Discovery and Beyond*, 2023
- Accelerating Computing for Emerging Sciences (ACES) Workshop, 2023
- SIAM Conference on Parallel Processing for Scientific Computing (PP22), 2022
- SIAM Conference on Computational Science and Engineering, attendee and poster judge
- Winter School, *Advanced Numerical Methods for Hyperbolic PDE*, Laboratory of Applied Mathematics, University of Trento, Italy, 2021

Selected Conference Posters

- *ICESEE: An Open-Source, Scalable Data Assimilation Library for Ice Sheet Models*, AGU 2025
- *GeoFlood: A Computational Model for Overland Flooding*, 2023 Graduate Showcase, Boise State University
- *GPU-Accelerated Wave Propagation Algorithm*, 2023 Research Computing Days Conference
- *Geothermal Closed Loop*, 2022 Clean Energy Development, Idaho National Laboratory
- *GPU-Accelerated Wave Propagation Algorithm*, 2022 3rd Biennial Meeting of the SIAM Pacific Northwest Section

Invited Talks and Visits

- ESCO 2026: *Advanced Computational Methods for Earth System Modeling and Analysis*
- *ICESEE: An Open-Source, Scalable Data Assimilation Framework for Ice Sheet Modeling*, 5th Biennial SIAM North Pacific Meeting
- *Math on ICE: ICESEE Coupling with ISSM for EnKF Data Assimilation*
- Seminar for the Geosystems Group: *Unified Computational Approaches to Flood and Ice Sheet Modeling*
- *Data Assimilation with Ice Sheet Modeling*, ART@SC24
- *GeoFlood: Computational Model for Overland Flooding*, 4th Biennial Meeting of the SIAM Pacific Northwest Section
- *GeoFlood: A Computational Model for Overland Flooding*, 2023 Lightning Talk, ACES Workshop
- Research Visit at USGS: *Overland Flow Modeling with David L. George*, June 2023
- Keynote Speaker, GAIN: Promoting and encouraging study abroad for students from low-developing countries

Activities, Associations, and Awards

- 5th Biennial SIAM North Pacific Meeting – Mini-symposium organizer and poster judge, *Data Assimilation for Earth and Atmospheric Systems* (October 2025)
- Georgia Institute of Technology – Research Faculty Advisory Council Representative for the School of Earth & Atmospheric Sciences (April 2025–Present)
- Vanderbilt University – Adviser & Icepack Workshop and Hackathon Grand Prize Winner
- Boise State University – NSBE Public Policy SIG STEM Education, Team Co-Lead (Oct. 2021 – Oct. 2022)
- Boise State University – NSBE Student Chapter, President (April 2023 – April 2024)
- Boise State University – NSBE Student Chapter, Vice President (April 2021 – April 2023)
- Boise State University – SIAM Student Chapter, President (Mar. 2022 – April 2023)
- 2023 Research Computing Days Hackathon – Organizer and Judge
- Idaho school outreach activities promoting STEM education
- Boise State University – SIAM Student Chapter, Vice President (Mar. 2021 – Mar. 2022)
- 2021 SIAM Travel Award (AN21)

- SIAM Travel Award (GS21)
- Graduate Research Assistantship Scholarship (Ph.D.), Aug. 2020 – July 2024
- 2023 Innovation Award Finalist, Promoting STEM Across Idaho
- Research Computing Days Conference – Winning Poster, *GPU Wave Propagation Algorithm*
- Association of State Floodplain Managers (ASFPM)
- United States Research Software Engineers (US-RSE)
- US-RSE 2023 Conference Travel Award
- AIMS Rwanda – AIMS Master’s Scholarship (M.Sc.), Aug. 2019 – July 2020
- ESMT Berlin – Team Lead (May 2020 – July 2020)
- Makerere University Physics Society – Vice President (April 2017 – April 2018)
- Government Scholarship on National Merit (B.Sc.), Aug. 2015 – May 2018

References

- Prof. Alexander Robel, Georgia Institute of Technology – alexander.robel@eas.gatech.edu
- Prof. Donna Calhoun, Boise State University – donnacalhoun@boisestate.edu
- Mr. Craig Earley, University of Colorado Boulder – craig.Earley@colorado.edu
- Dr. David L. George, U.S. Geological Survey – dgeorge@usgs.gov